

Node.js Interview Revision Sheet (English + Hinglish Mix)

1■■■ Core Concepts

Event Loop: Manages async tasks in Node.js. It keeps checking callback queue & executes when main thread is free. Think of it as the manager of all background tasks.

Non-Blocking I/O: Node doesn't wait for a task to finish. It moves to the next task while previous ones run in the background.

2■■■ Async Handling

Callback: Function inside another function, executed after task completion.

■ Interview line: 'Callback allows async operations to run after main task finishes.'

Promise: Represents a future result (resolve/reject).

■ Interview line: 'Promise handles async operations with success or failure states.'

Async/Await: Cleaner syntax for Promises, makes async code look sync.

■ Interview line: 'Async/Await simplifies async code readability.'

3■■■ Express.js

Middleware: Functions that process requests before reaching routes (like logger, auth, etc.).

Routing: Defines endpoints for API (GET, POST, PUT, DELETE).

Error Handling: Use try-catch or custom middleware to handle errors gracefully.

4■■■ REST API Structure

Follow MVC pattern: routes → controllers → models.

Use proper HTTP methods (GET, POST, PUT, DELETE).

Return JSON response with status codes (200, 400, 404, 500).

5■■■ MongoDB + JWT

MongoDB: NoSQL database; use Mongoose for schema & CRUD operations.

JWT: Used for authentication; token created at login & verified in protected routes.

6■■■ Common Interview Questions

- What is Event Loop and how does it work?
- Difference between blocking and non-blocking?
- How does middleware work in Express?
- What are Promises and why use them?
- How to handle errors in async functions?
- How JWT authentication works?

■ Tip: Be confident, explain with small code examples, and focus on logic clarity. Webs Optimization values practical knowledge more than deep theory.